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A				FORCED / INDUCED DRAFT (FD/ID) FANS MAINTENANCE FRAGNET Routine/Weekly Checks Lubrication: Verify grease/lube oil levels, operating temperatures, and flow to bearings. Vibration & Noise: Monitor for unusual noises or vibrations Temperature: Check bearing and motor winding temperatures. Leaks: Visually inspect for air, oil, or cooling water leaks. Monthly/Detailed Checks Impeller & Blades: Clean accumulated dust and inspect for pitting, erosion, or corrosion. Belts & Couplings: Inspect V-belts for wear and correct tension. Check coupling alignment and bolt tightness. Fasteners & Mountings: Tighten foundation bolts and inspect static parts (ducts, inlet cones). TA/Shutdown Overhaul Clearance Check: Verify the clearance between the inlet cone and the impeller using inside calipers. Shaft & Bearings: Inspect shaft seals. Measure bearing clearances and replace if worn. Coupling/Motor: Decouple the motor to check base bolt locking and perform alignment checks. Dampers: Check the open/close operation of suction/discharge dampers and Inlet Guide Vanes (IGV). Body corrossion check and refurbishment Bonnets painting or replacement Shafts alignment checks Bearing and couplings change Internal checks on dampers and cleaningg of deposited soot and other choking materials Belts replacement Motors function checks or rewinding Cabling and connection continuity checks																										
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D	ID FAN & FD FAN IN CHEMICAL INDUSTRIES (EQUIPMENT DIAGRAM & WORKING) FD FAN (Forced Draft Fan) FUNCTION FD Fan supplies fresh air to the furnace for proper combustion. WORKING It draws atmospheric air. Increases the air pressure. Pushes the air through air preheater and into the furnace. Maintains Positive Pressure in furnace. APPLICATION Boilers Furnaces Heaters Incinerators AIR FLOW PATH 1. Atmospheric Air 2. FD Fan (Forced Draft Fan) 3. Air Preheater (APH) 4. Air Duct 5. Air Inlet to Furnace KEY POINTS Provides air for combustion Increases pressure Positive pressure in furnace Important for efficient combustion FD FAN SUPPLIES AIR (Positive Pressure) AIR FLOW (FD FAN SIDE) AIR INLET TO FURNACE AIR PREHEATER (APH) FURNACE / COMBUSTION CHAMBER FLUE GAS FLOW FLUE GAS FLOW (ID FAN SIDE) ECONOMIZER AIR PREHEATER (APH) GAS DUCT ID FAN DRAWN FLUE GAS (Negative Pressure) STACK / CHIMNEY KEY POINTS Removes flue gases Creates suction Negative pressure in furnace Improves efficiency & safety IMPORTANT NOTE Both fans are essential in any combustion system. FD Fan ensures sufficient air for burning the fuel and ID Fan ensures gaseous products are safely removed and system operates efficiently. COMPARISON BETWEEN ID FAN & FD FAN <table><thead><tr><th>PARAMETER</th><th>FD FAN (Forced Draft Fan)</th><th>ID FAN (Induced Draft Fan)</th></tr></thead><tbody><tr><td>Function</td><td>Supplies fresh air for combustion</td><td>Removes flue gases from system</td></tr><tr><td>Pressure</td><td>Positive Pressure</td><td>Negative Pressure (Suction)</td></tr><tr><td>Location</td><td>At air inlet side (Before furnace)</td><td>At gas outlet side (After furnace)</td></tr><tr><td>Flow Direction</td><td>Air flows into furnace</td><td>Flue gas flows out of furnace</td></tr><tr><td>Purpose</td><td>To support combustion</td><td>To maintain suction & remove gases</td></tr><tr><td>Typical Use</td><td>Boilers, Heaters, Furnaces</td><td>Boilers, Kilns, Process Units</td></tr></tbody></table>				PARAMETER	FD FAN (Forced Draft Fan)	ID FAN (Induced Draft Fan)	Function	Supplies fresh air for combustion	Removes flue gases from system	Pressure	Positive Pressure	Negative Pressure (Suction)	Location	At air inlet side (Before furnace)	At gas outlet side (After furnace)	Flow Direction	Air flows into furnace	Flue gas flows out of furnace	Purpose	To support combustion	To maintain suction & remove gases	Typical Use	Boilers, Heaters, Furnaces	Boilers, Kilns, Process Units	Drw Purpose Code :- Status of Approval :- Dimentions as Mentioned Drawing Scale :- Engg Code :- Backup Info Code :- Angle of Projection Sheet Size			Statutory Note: This document is prepared by rPlan and is the sole proprietary of the company. Its purpose is educational and no references within this document are for Engineering or Design correlation with the applicable codes. For any clarifications beyond the intended purpose viewers can always correspond to iPlan via its company website. This drw is not for duplication without company permission! Drw electronic code rPlan AT 39, MERMAID 1 CHS LTD., SEC-11, CBD-BELAPUR, NAVI MUMBAI - 400614 Legal Owner QR CODE	Description: Typical FANS Fragnet Preliminary Doc under prep Drawn By: Drawn Date Checked Approved by Checked Approved Date Drawing Number Sheet Rev 1 of 1 XX
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